

CYBR 437 Secure Coding

Fall 2026

Lab-1

Release date: September 24, 2026

Due date: September 30, 2026, at 11:59 pm

Introduction

In this lab, you will modify/write a few C programs, execute them, and analyze the outputs. You will use your individual Ubuntu Linux virtual machine assigned for the labs in this course. Your VM's IP address, username, and password have been provided to you through a Canvas message.

You will SSH into the VM from Terminal/Command Prompt/Windows PowerShell with your username (for example, `ssh aborah@10.202.12.48`) and enter your password when prompted. After logging in for the first time, please change your password when prompted.

We will use the standard C compiler for Ubuntu, the GNU Compiler Collection (GCC), in this course. For a quick review of C, you may find the following links useful (of course, you can refer to your preferred C language resources that you might have used before): <https://www.w3schools.com/c/> or <https://www.tutorialspoint.com/cprogramming/index.htm>.

Tasks:

1. (15 points) Examine the following program and perform the tasks given on the next page.

```
#include <stdio.h>

int main()
{
    char c = 'C';
    char *p = &c;

    printf("p is %p\n", p);
    printf("The value at p is %c\n", *p);

    /* Pointer arithmetic to find what p+1 is */
    p = p + 1;
    printf("Now p is %p\n", p);

    return 0;
}
```

- a) (3 points) Execute the program and explain the output.
 - b) (3 points) Modify the code to perform the same pointer arithmetic on a pointer to an int. Execute the program and explain the output.
 - c) (3 points) Modify the code again to perform the same pointer arithmetic on a pointer to a double. Execute the program and explain the output.
 - d) (6 points) What should happen if the line $p = p + 1$ is changed to $p = p + 2$ in parts a-c above? Execute the program with this change (for all of parts a-c) to verify your answer.
2. (5 points) In the following program, add the required lines of code to print the value and address of variable x in `fun1`, and variable y in `fun2`. Execute the program and precisely explain the output.

```
#include <stdio.h>

void fun1(int xval)
{
    int x;
    x = xval;
    /* print the address and value of x here */

}

void fun2()
{
    int y = 10;
    /* print the address and value of y here */

}

int main()
{
    fun1(5);
    fun2();
    return 0;
}
```

3. (5 points) Write a program that declares and initializes (to any value) a double and an int. Your program should then print the address and the value stored in each of the variables, along with the amount of memory each variable occupies.

4. (5 points) Write a function that accepts two double variables as parameters and swaps their values. Then call the function in the main function to verify that your function works correctly.

Submission Instructions: Please submit a single PDF file in Canvas that includes:

1. C code and/or explanations for all the tasks in text format (please do not include the code as screenshots).
2. Screenshots showing the successful execution and output of each program.